Solution Requirements

Date	02 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Allow authorized bank staff to securely access the application.	High
2	Authorization levels	Restrict prediction and applicant management features to authorized users.	High

3	External interfaces	Read applicant and credit history datasets (application_record.csv and credit_record.csv) and interact with the trained ML model through the Flask application.	**High**
4	Transactions processing	Accept applicant details, preprocess the data, generate a prediction using the trained model, and return the approval/rejection result.	**High**
5	Reporting	Display the prediction result along with the risk category through the web interface.	**Medium**
6	Business rules, etc.	Perform duplicate removal, handle missing values, encode categorical features, merge datasets, and classify applicants based on the trained machine learning model.	**High**
7	Compliance to laws or regulations	Protect applicant information and use the dataset only for educational and research purposes while maintaining data privacy.	**Medium**
8	*Other*	Support multiple machine learning algorithms (Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting/XGBoost) and allow selection of the best-performing model.	**Medium**

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Generate prediction quickly after user input.	Prediction displayed within **2–3 seconds**.
2	Scalability	Support prediction requests for multiple applicants without significant performance degradation.	Able to process hundreds of prediction requests efficiently.
3	Security & Data Privacy	Protect applicant information and restrict unauthorized access to the system.	Secure handling of user data and authenticated access
4	Reliability & Availability	Ensure the prediction service remains available during normal usage.	**99% availability** during project demonstration/testing.
5	Usability & Accessibility	Provide a simple, user-friendly web interface that is easy for bank staff to use.	Easy navigation with clear input forms and prediction results.
6	*Other*	Write modular, well-documented code to simplify updates and future enhancements.	Easy to maintain and extend with new ML models or features.